BP105T -Introduction to Pharmacognosy (Theory)

UNIT-1

Fundamentals of Pharmacognosy (a) Definition, history, present status, scope and development of pharmacognosy. (b) Sources of drugs: Plants, animals, microbial, marine, mineral and plant tissue culture. (c) Historical milestones in drug discovery: Morphine, quinine, aspirin, warfarin, penicillin, cephalosporin, taxol and artemisinin. (d) Introduction to different herbal / traditional pharmacopoeias: Indian Pharmacopoeia, British Herbal Pharmacopoeia, United States Pharmacopeia – Herbal Medicines and Dietary Supplements, Ayurvedic Pharmacopoeia of India, Unani Pharmacopoeia of India and American Herbal Pharmacopoeia. (e) Official and non-official; codified and non-codified drugs. Classification of crude drugs: alphabetical, morphological, taxonomical, chemical, pharmacological and chemotaxonomic classification along with their merits and limitations.

FUNDAMENTALS OF PHARMACOGNOSY

- Pharmacognosy is the branch of pharmaceutical science that deals with the study of natural drugs obtained from plants, animals, microorganisms, and minerals.
- It Includes The Identification, Collection, Cultivation, processing, evaluation, and medicinal uses of crude drugs.
- Pharmacognosy plays an important role in the discovery and development of safe and effective medicines from natural sources.

DEFINITION, HISTORY, PRESENT STATUS, SCOPE AND DEVELOPMENT OF PHARMACOGNOSY.

Definition of Pharmacognosy

Pharmacognosy is the branch of pharmaceutical science that deals with the study of crude drugs obtained from natural sources such as plants, animals, microorganisms, and minerals, including their identification, cultivation, collection, processing, evaluation, and medicinal uses.

Word Origin:

- Pharmakon = Drug (Greek)
- Gnosis = Knowledge

Pharmacognosy = Knowledge of Drugs

History of Pharmacognosy

Ancient Period

- Medicinal plants were used in India, China, Egypt, and Greece for thousands of years.
- Indian systems like Ayurveda described many herbal medicines.

Greek Period

- Dioscorides wrote the famous book "De Materia Medica", describing about 600 medicinal plants.
- He is known as the Father of Pharmacognosy.

19th Century

- The term "Pharmacognosy" was first used by C.A. Seydler in 1815.
- Scientific methods for identification and evaluation of crude drugs were introduced.

Modern Period

- Pharmacognosy developed with advances in phytochemistry, biotechnology, and quality control.
- It plays an important role in herbal drug research and the discovery of new medicines.

PRESENT STATUS OF PHARMACOGNOSY

Today, pharmacognosy is an important branch of pharmacy because:

- Increasing use of herbal medicines worldwide.
- Discovery of new drugs from natural sources.
- Standardization and quality control of herbal products.
- Use of modern techniques such as chromatography and spectroscopy.
- Growing demand for safe, natural, and plant-based medicines.

SCOPE OF PHARMACOGNOSY

The scope of pharmacognosy includes:

- Identification of crude drugs.
- Cultivation and collection of medicinal plants.
- Processing and storage of crude drugs.
- Evaluation and quality control of herbal drugs.
- Isolation of active constituents.
- Development of herbal formulations.
- Discovery of new drugs from natural sources.
- Research in phytochemistry and biotechnology.

DEVELOPMENT OF PHARMACOGNOSY

The development of pharmacognosy has progressed through the following stages:

1. Traditional use of medicinal plants.
2. Scientific identification of crude drugs.
3. Isolation of active chemical constituents.
4. Standardization and quality control.
5. Application of biotechnology and molecular biology.
6. Development of modern herbal medicines and nutraceuticals.

Sources of drugs: Plants, animals, microbial, marine, mineral and plant tissue culture.

1. Plant Sources

Plant sources are the major source of natural drugs. Drugs are obtained from different parts of plants such as roots, stems, leaves, flowers, fruits, seeds, and bark. They contain active constituents used in the treatment of various diseases.

Examples: Digitalis, Aloe, Neem, Turmeric, Cinchona

Advantages

- Easily available
- Economical
- Fewer side effects

Disadvantages

- Seasonal variation
- Adulteration
- Slow growth

2. Animal Sources

Animal drugs are obtained from animal organs, glands, tissues, blood, or secretions. They are mainly used for producing hormones, enzymes, and biological products. These drugs play an important role in modern medicine.

Examples: Insulin, Heparin, Cod liver oil, Honey

Advantages

- Highly effective
- Source of hormones & enzymes
- Used in life-saving therapies

Disadvantages

- Disease transmission
- Ethical issues
- Expensive

3. Microbial Sources

Microbial drugs are obtained from bacteria, fungi, and other microorganisms. They are mainly produced by fermentation and are an important source of antibiotics, vaccines, and enzymes.

Examples: Penicillin, Streptomycin, Cephalosporin

Advantages

- Fast production
- Large-scale manufacturing
- Important source of antibiotics

Disadvantages

- Contamination risk
- Drug resistance
- Requires sterile conditions

4. Marine Sources

Marine drugs are obtained from sea plants and marine animals such as algae, sponges, and fish. They contain unique bioactive compounds useful in treating various diseases. Marine organisms are also important in drug discovery research.

Examples: Agar, Carrageenan, Fish oil (Omega-3)

Advantages

- Rich in bioactive compounds
- Useful in cancer research
- High medicinal value

Disadvantages

- Difficult collection
- Costly
- Marine pollution affects quality

5. Mineral Sources

Mineral drugs are obtained from naturally occurring minerals and are purified before medicinal use. They are commonly used as antacids, laxatives, and pharmaceutical excipients due to their stability.

Examples: Kaolin, Talc, Calcium carbonate, Magnesium sulfate

Advantages

- Stable
- Long shelf life
- Low cost

Disadvantages

- Some are toxic
- Purification required
- Limited therapeutic use

6. Plant Tissue Culture

Plant tissue culture is the in vitro cultivation of plant cells, tissues, or organs on a sterile nutrient medium. It is used for the large-scale production of medicinal compounds and conservation of rare medicinal plants.

Examples: Taxol, Atropine, Shikonin

Advantages

- Disease-free plants
- Rapid production
- Conserves rare plants

Disadvantages

- Expensive
- Skilled personnel required
- Specialized laboratory needed

SYPT
STUDY YOURSELF PHARMACY TOGETHER
Gain Knowledge Every Day. Believe in You, Keep Growing. Delivering Care, Improving Lives. Learn, Share & Achieve.
PHARMACY NOTES
Smart Notes. Strong Concepts. Bright Future.
LEARN REVISE APPLY ACHIEVE

SOURCES OF DRUGS – QUICK RECALL

1. PLANT SOURCES



Explanation (3–5 lines)

Plant sources are the major source of natural drugs. Drugs are obtained from different parts like roots, stems, leaves, flowers, fruits, seeds, and bark. They contain active constituents used in the treatment of various diseases.

Examples

- Digitalis
- Aloe
- Neem
- Turmeric
- Cinchona

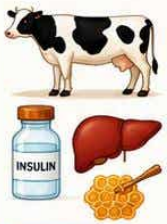
Advantages

- ✓ Easily available
- ✓ Economical
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Disadvantages

- ✗ Seasonal variation
- ✗ Adulteration
- ✗ Slow growth

2. ANIMAL SOURCES



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Animal drugs are obtained from animal organs, glands, tissues, blood, or secretions. They are mainly used for producing hormones, enzymes, and biological products. These drugs play an important role in modern medicine.

Examples

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- Heparin
- Cod liver oil
- Honey

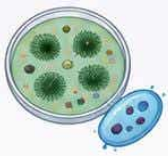
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- ✓ Highly effective
- ✓ Source of hormones & enzymes
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Disadvantages

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- ✗ Ethical issues
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- Streptomycin
- Cephalosporin

Advantages

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- ✓ Large-scale manufacturing
- ✓ Important source of antibiotics

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- ✗ Drug resistance
- ✗ Requires sterile conditions

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MEMORY TRICK

P	A	M	M	M	P
Plants	Animals	Microbial	Marine	Mineral	Plant Tissue Culture

QUICK FLOW

PLANTS → ANIMALS → MICROBIAL → MARINE → MINERAL → PLANT TISSUE CULTURE

HISTORICAL MILESTONES IN DRUG DISCOVERY

Drug discovery is the process of identifying and developing new medicines for the prevention and treatment of diseases. From ancient herbal remedies to modern biotechnology, several important milestones have contributed to the development of effective drugs.

Historical Milestones

1. Ancient Period (3000–1500 BC)

- Medicinal plants were used in India, China, and Egypt.
- Traditional systems like Ayurveda and Traditional Chinese Medicine (TCM) were developed.
- Herbs were used to treat various diseases naturally.

2. Greek Period (1st Century AD)

- Dioscorides wrote "De Materia Medica".
- The book described about 600 medicinal plants.
- It became the foundation of Pharmacognosy.

3. 1815 – Introduction of Pharmacognosy

- C.A. Seydler first used the term "Pharmacognosy".
- Scientific study of crude drugs began.

4. 1806 – Isolation of Morphine

- Friedrich Sertürner isolated Morphine from opium.
- It was the first pure alkaloid isolated from a plant.

5. 1928 – Discovery of Penicillin

- Alexander Fleming discovered Penicillin from *Penicillium* fungus.
- It marked the beginning of the antibiotic era.

6. 1953 – DNA Structure

- James Watson and Francis Crick discovered the double-helix structure of DNA.
- This revolutionized genetics and drug research.

7. Modern Era

- Drug discovery now uses biotechnology, genetic engineering, computer-aided drug design, and artificial intelligence (AI).
- Modern techniques help develop safer and more effective medicines.

1. Morphine (1806)

Morphine is a natural alkaloid isolated from the dried latex (opium) of *Papaver somniferum*. It was first isolated by Friedrich Sertürner in

1806 and is considered the first pure alkaloid obtained from a medicinal plant. It is one of the most effective drugs for relieving severe pain**.**

Source

- **Plant:** *Papaver somniferum* (Opium poppy)
- **Part Used:** Dried latex (Opium)

Importance

- First pure alkaloid isolated from a plant.
- Powerful opioid analgesic.
- Used for severe and cancer pain.
- Used after major surgeries.
- Milestone in alkaloid chemistry.
- Helped in modern drug research.

2. Quinine (1820)

Quinine is a natural alkaloid isolated from the bark of the Cinchona tree. It was discovered by Pelletier and Caventou in 1820. It became

the first effective medicine for the treatment of malaria and played a major role in controlling the disease.

Source

- **Plant:** *Cinchona* species

- **Part Used:** Bark

Importance

- First effective antimalarial drug.
- Treats malaria.
- Opened alkaloid research.
- Reduced malaria deaths.
- Important milestone in tropical medicine.
- Used as a standard antimalarial for many years.

3. Aspirin (1897)

Aspirin was developed by Felix Hoffmann in 1897 from salicin obtained from willow bark. It is one of the most widely used medicines and is effective in relieving pain, reducing fever, and controlling inflammation.

Source

- **Plant:** Willow (*Salix* species)
- **Active Compound:** Salicin

Importance

- Relieves pain.
- Reduces fever.
- Anti-inflammatory action.

- Prevents blood clot formation (low dose).
- Widely used worldwide.
- One of the oldest and safest medicines.

4. Warfarin (1948)

Warfarin is an anticoagulant drug developed from dicoumarol found in spoiled sweet clover. It is used to prevent the formation of harmful blood clots in patients with cardiovascular diseases.

Source

- **Plant:** Sweet Clover (*Melilotus* species)

Importance

- Prevents blood clotting.
- Reduces risk of stroke.
- Used in heart diseases.
- Prevents deep vein thrombosis (DVT).
- Life-saving anticoagulant.
- Used for long-term therapy.

5. Penicillin (1928)

Penicillin was discovered accidentally by Alexander Fleming in 1928 from the fungus *Penicillium notatum*. It was the first antibiotic discovered and revolutionized the treatment of bacterial infections.

Source

- **Microorganism:** *Penicillium notatum* (Fungus)

Importance

- First antibiotic.
- Treats bacterial infections.
- Reduced infection-related deaths.
- Revolutionized modern medicine.
- Opened the antibiotic era.
- Saved millions of lives.

6. Cephalosporin (1945)

Cephalosporin is a broad-spectrum antibiotic discovered by Giuseppe Brotzu from the fungus *Cephalosporium acremonium*. It is

effective against a wide variety of bacterial infections and is often used when penicillin cannot be given.

Source

- **Microorganism:** *Cephalosporium acremonium* (*Acremonium*)

Importance

- Broad-spectrum antibiotic.
- Effective against Gram-positive and Gram-negative bacteria.
- Alternative to penicillin.

- Treats severe infections.
- Lower allergy risk than penicillin (some cases).
- Widely used in hospitals.

7. Taxol (Paclitaxel) (1967)

Taxol is a natural anticancer drug isolated from the bark of the Pacific yew tree (*Taxus brevifolia*). It prevents cancer cells from dividing and is widely used in chemotherapy.

Source

- **Plant:** *Taxus brevifolia* (Pacific yew)
- **Part Used:** Bark

Importance

- Anticancer drug.
- Used in breast cancer.
- Used in ovarian cancer.
- Used in lung cancer.
- Prevents cancer cell division.
- Improved cancer survival rates.

8. Artemisinin (1972)

Artemisinin is a natural antimalarial compound isolated from *Artemisia annua*. It was discovered by Tu Youyou in 1972 and is one of the most effective drugs for treating malaria, especially drug-resistant malaria.

Source

- **Plant:** *Artemisia annua*
- **Part Used:** Leaves and flowering tops

Importance

- Highly effective against malaria.
- Works against drug-resistant malaria.
- Rapid action.
- Saves millions of lives.
- Basis of ACT (Artemisinin Combination Therapy).
- Tu Youyou received the Nobel Prize (2015).

INTRODUCTION TO DIFFERENT HERBAL / TRADITIONAL PHARMACOPOEIAS

Herbal or traditional pharmacopoeias are official reference books that contain standards and detailed information about medicinal plants, crude drugs, and herbal formulations. They provide guidelines for the identification, collection, preparation, storage, quality control, purity, strength, and therapeutic uses of herbal medicines. These pharmacopoeias ensure that herbal drugs are safe, effective, and of good quality. They also help pharmacists, researchers, and manufacturers maintain uniform standards for herbal medicines used in healthcare.

Key Points

- Official reference books for herbal medicines.
- Describe medicinal plants and crude drugs.

- Provide standards for identity, purity, quality, and strength.
- Help in the preparation and quality control of herbal drugs.
- Ensure the safety, efficacy, and standardization of traditional medicines.

1. Indian Pharmacopoeia (IP)

Indian Pharmacopoeia (IP) is the official book of drug standards in India. It contains monographs and quality standards for pharmaceutical substances, dosage forms, and selected herbal drugs. It is published by the Indian Pharmacopoeia Commission (IPC), Ministry of Health and Family Welfare, Government of India.

Principle

- ✓ The Indian Pharmacopoeia is based on the principle of standardization and quality assurance of medicines.
- ✓ It ensures that every drug meets the required standards of identity, purity, strength, quality, and safety before reaching patients.
- ✓ It provides official analytical methods for testing drugs and detecting impurities.
- ✓ The standards are revised regularly according to scientific advancements.
- ✓ Thus, it ensures the production of safe, effective, and uniform quality medicines.

Features

- Official drug standard book of India.
- Published by Indian Pharmacopoeia Commission (IPC).
- Contains drug monographs and analytical methods.
- Provides standards for purity, strength, and quality.
- Includes storage, packaging, and labeling requirements.
- Revised periodically.

Advantages

- Ensures uniform drug quality.
- Improves patient safety.
- Prevents adulteration.
- Helps in quality control.
- Accepted by pharmaceutical industries and regulatory authorities.

Disadvantages

- Requires frequent revisions.
- Implementation may be costly for small manufacturers.
- Covers limited traditional herbal medicines compared to Ayurvedic pharmacopoeia.

Importance

- Protects public health.
- Maintains drug quality standards.
- Guides pharmaceutical manufacturers.
- Supports drug regulation.
- Ensures safe and effective medicines.

2. British Herbal Pharmacopoeia (BHP)

Principle

- ✓ The British Herbal Pharmacopoeia aims to standardize herbal medicines by providing scientific information on medicinal plants.
- ✓ It ensures the correct identification, purity, quality, dosage, and therapeutic uses of herbal drugs.
- ✓ It also includes precautions and contraindications for safe use.
- ✓ The pharmacopoeia promotes evidence-based herbal medicine and helps maintain consistency in herbal products.

Advantages

- Standardizes herbal medicines.
- Provides scientific information.
- Improves quality control.
- Supports herbal practitioners.

Disadvantages

- Mainly applicable to the UK.
- Limited coverage of non-European medicinal plants.
- Needs periodic updates.

Importance

- Ensures safe herbal medicines.
- Supports herbal research.
- Improves product quality.
- Promotes rational use of herbs.
- Used as an official reference.

3. USP – Herbal Medicines & Dietary Supplements

Principle

- ✓ The USP establishes scientifically validated standards for herbal medicines and dietary supplements.
- ✓ It ensures that products meet the required standards of identity, purity, strength, quality, and performance.
- ✓ It also provides validated analytical methods and quality testing procedures.
- ✓ These standards help protect consumers from adulterated or substandard products and improve confidence in herbal medicines.

Advantages

- Internationally recognized.
- Reliable scientific standards.
- Prevents adulteration.

- Ensures product quality.

Disadvantages

- Expensive quality testing.
- Complex analytical procedures.
- Mainly focused on the US market.

Importance

- Protects consumer health.
- Supports global trade.
- Maintains international quality standards.
- Improves product reliability.
- Encourages scientific evaluation.

4. Ayurvedic Pharmacopoeia of India (API)

Principle

- ✓ The Ayurvedic Pharmacopoeia of India is based on the principle of standardizing Ayurvedic medicines according to classical Ayurvedic texts and modern scientific methods.
- ✓ It ensures the correct identification, purity, quality, and therapeutic efficacy of medicinal plants and formulations.
- ✓ It also provides official testing methods for quality assurance.
- ✓ This helps preserve the authenticity of Ayurvedic medicines while ensuring their safe and effective use.

Advantages

- Preserves Ayurvedic knowledge.
- Ensures quality of Ayurvedic drugs.
- Promotes safe herbal medicines.
- Supports research and standardization.

Disadvantages

- Limited to Ayurvedic medicines.
- Some traditional formulations require more scientific validation.

- Needs continuous revision.

Importance

- Standardizes Ayurvedic drugs.
- Protects traditional medicine.
- Improves global acceptance of Ayurveda.
- Helps manufacturers maintain quality.
- Ensures safe and effective treatment.

5. Unani Pharmacopoeia of India (UPI)

The Unani Pharmacopoeia of India (UPI) is the official pharmacopoeia for Unani medicines in India. It is published by the Ministry of AYUSH, Government of India, and provides quality standards for Unani crude drugs and formulations.

Principle

- ✓ The Unani Pharmacopoeia is based on the principle of standardization and quality assurance of Unani medicines.
- ✓ It ensures the correct identification, purity, quality, strength, and safety of crude drugs and formulations.
- ✓ It provides official methods for testing and evaluating Unani drugs.
- ✓ It also helps preserve the traditional Unani system of medicine while maintaining scientific standards.
- ✓ Thus, it ensures safe and effective use of Unani medicines.

Features

- Official pharmacopoeia for Unani medicines.
- Published by the Ministry of AYUSH.
- Contains monographs of Unani crude drugs.
- Provides standards for identification and purity.
- Includes quality control and analytical methods.
- Revised periodically.

Advantages

- Standardizes Unani medicines.
- Ensures quality and safety.
- Preserves traditional knowledge.
- Helps manufacturers and researchers.
- Promotes global acceptance of Unani medicine.

Disadvantages

- Limited only to Unani medicines.
- Some traditional formulations need more scientific validation.
- Requires regular updates.

Importance

- Maintains quality standards.
- Protects consumers.
- Supports quality control.
- Encourages research.
- Promotes safe use of Unani medicines.

6. American Herbal Pharmacopoeia (AHP)

The American Herbal Pharmacopoeia (AHP) is an independent scientific organization that publishes monographs and quality standards for medicinal plants and herbal products used in the United States.

Principle

- ✓ The American Herbal Pharmacopoeia is based on the principle of scientific standardization of herbal medicines.
- ✓ It provides accurate information on the identity, purity, quality, safety, and therapeutic uses of medicinal plants.
- ✓ It includes validated analytical methods for quality evaluation and promotes evidence-based use of herbal products.

✓ It also supports the production of safe and effective herbal medicines.

Features

- Publishes scientific herbal monographs.
- Provides quality standards for herbal drugs.
- Includes botanical identification methods.
- Describes analytical and quality control techniques.
- Gives information on dosage and safety.
- Widely used by researchers and industries.

Advantages

- Scientifically reliable.
- Improves herbal drug quality.
- Supports research and education.
- Promotes safe use of herbal medicines.
- Internationally recognized.

Disadvantages

- Mainly focused on herbal medicines.
- Some monographs require frequent updates.
- Less emphasis on traditional medicine systems.

Importance

- Ensures quality of herbal products.
- Prevents adulteration.
- Supports evidence-based herbal medicine.
- Helps manufacturers and pharmacists.
- Promotes international standardization.

OFFICIAL AND NON-OFFICIAL; CODIFIED AND NON-CODIFIED DRUGS.

Drugs are classified based on their recognition in pharmacopoeias and traditional literature. They are mainly classified as Official & Non-official drugs and Codified & Non-codified drugs.

1. Official Drugs

Official drugs are those drugs that are included in an official pharmacopoeia or formulary and have prescribed standards for identity, purity, quality, and strength.

Examples

- Aspirin (IP)
- Paracetamol (IP)
- Digitalis (IP)

Advantages

- Standard quality.
- Safe and effective.
- Legally recognized.
- Easy quality control.

Disadvantages

- Regular revision required.
- New drugs may not be included immediately.

Importance

- Ensures drug quality.
- Protects public health.
- Maintains uniform standards.

2. Non-Official Drugs

Non-official drugs are those drugs that are not included in any official pharmacopoeia or formulary but are still used in medical practice or research.

Examples

- Some newly discovered herbal drugs.
- Experimental drugs.

Advantages

- Encourages innovation.
- Useful in research.
- May provide new therapeutic options.

Disadvantages

- No official quality standards.
- Safety and efficacy may not be fully established.

Importance

- Source of future medicines.
- Helps in drug discovery.

3. Codified Drugs

Codified drugs are drugs that are described in traditional medical texts or official pharmacopoeias with proper standards, methods of preparation, dosage, and therapeutic uses.

Examples

- Ashwagandha
- Tulsi
- Neem
- Triphala

Advantages

- Scientifically documented.
- Standard methods available.

- Easy identification.
- Reliable quality.

Disadvantages

- Limited flexibility.
- Requires periodic revision.

Importance

- Preserves traditional knowledge.
- Ensures quality and uniformity.
- Promotes safe use.

4. Non-Codified Drugs

Non-codified drugs are medicinal substances that are not described in official pharmacopoeias or classical traditional texts. They are mainly used in folk medicine or local traditional practices.

Examples

- Local medicinal herbs.
- Newly identified medicinal plants.

Advantages

- Source of new drug discovery.
- Preserves local traditional knowledge.
- Easily available in rural areas.

Disadvantages

- No standard quality.
- Limited scientific evidence.
- Safety and dosage are uncertain.

Importance

- Provides leads for research.
- Expands medicinal plant knowledge.
- May become future codified drugs.

CLASSIFICATION OF CRUDE DRUGS

Classification of crude drugs is the systematic arrangement of drugs into different groups based on their common characteristics such as name,morphology,taxonomy, chemical constituents,pharmacological action,chemotherapeutic relationship. It helps in easy identification, study, evaluation, and proper use of crude drugs.

1. Alphabetical Classification

- ✓ In alphabetical classification, crude drugs are arranged alphabetically according to their English, Latin, scientific, or vernacular names.
- ✓ It is the simplest and most commonly used method in pharmacopoeias, textbooks, and reference books.
- ✓ This method allows quick identification and easy location of a drug. However, it does not provide information about the source, chemical constituents, or medicinal action of the drug.

Merits (Advantages)

- Very simple and easy to understand.
- Easy to locate drugs quickly.
- Saves time in searching.
- Used in pharmacopoeias and dictionaries.
- No scientific knowledge required.
- Useful for beginners.
- Convenient for indexing and cataloging.

Limitations (Disadvantages)

- No information about chemical constituents.
- Does not indicate therapeutic action.
- Does not show botanical relationship.
- Similar drugs are separated.
- Not useful for research.
- No idea about morphology.

- Cannot compare related drugs.

2. Morphological Classification

✓ Morphological classification is based on the external appearance and the plant part used as the crude drug.

✓ Drugs are grouped according to roots, stems, leaves, flowers, fruits, seeds, bark, wood, or whole plants.

✓ It is useful for the identification of organized crude drugs using visible characters.

✓ This method is widely used in pharmacognosy laboratories.

Merits (Advantages)

- Easy identification.
- Based on visible characters.
- Useful for practical studies.
- Simple classification.
- Helps in drug collection.
- Useful for organized drugs.
- Easy to remember.

Limitations (Disadvantages)

- Not useful for unorganized drugs.
- No information on chemical constituents.
- No pharmacological classification.
- Similar plant parts may differ chemically.
- Cannot identify adulteration completely.
- No taxonomic relationship.
- Limited scientific value.

3. Taxonomical Classification

✓ Taxonomical classification groups crude drugs according to their botanical classification based on family, genus, and species.

- ✓ It follows the scientific system of plant classification and shows the natural relationship among plants.
- ✓ Drugs belonging to the same family usually possess similar characteristics.
- ✓ This method is widely used in botanical and pharmacognostic studies and helps in accurate identification of medicinal plants.

Merits (Advantages)

- Scientifically accepted.
- Shows natural relationship.
- Accurate identification.
- Useful in botanical studies.
- Helps identify related species.
- Reduces confusion.
- Useful for research.

Limitations (Disadvantages)

- Difficult for beginners.
- No information about medicinal action.
- No chemical classification.
- Time-consuming.
- Requires botanical knowledge.
- Not suitable for animal drugs.
- Limited practical use.

4. Chemical Classification

- ✓ Chemical classification groups crude drugs according to their major active chemical constituents such as alkaloids, glycosides, tannins, volatile oils, resins, flavonoids, and steroids.
- ✓ Drugs with similar chemical constituents are placed in the same group.
- ✓ This classification is useful for phytochemical studies, quality control, and isolation of active compounds.

✓ It is widely used in pharmaceutical industries and research laboratories.

Merits (Advantages)

- Scientifically reliable.
- Useful for phytochemistry.
- Helps in drug standardization.
- Easy isolation of active compounds.
- Useful for quality control.
- Supports research.
- Helps in drug development.

Limitations (Disadvantages)

- Requires chemical analysis.
- Expensive.
- Complex method.
- Some drugs contain multiple constituents.
- Not useful for beginners.
- Ignores botanical relationships.
- Needs laboratory facilities.

5. Pharmacological Classification

✓ Pharmacological classification groups crude drugs according to their therapeutic effects or pharmacological actions on the human body.

✓ Drugs with similar medicinal uses are placed in the same category, such as laxatives, analgesics, expectorants, or antihypertensives.

✓ This classification is very useful for physicians, pharmacists, and clinicians in selecting appropriate drugs for disease treatment.

Merits (Advantages)

- Easy for medical use.
- Based on therapeutic action.
- Helpful in treatment.
- Useful for pharmacists.
- Simple to understand.
- Helps in prescribing drugs.
- Clinically important.

Limitations (Disadvantages)

- One drug may have many actions.
- Similar action from different chemicals.
- No botanical relationship.
- No chemical information.
- Difficult to classify multi-purpose drugs.
- Not suitable for research.
- Limited scientific classification.

6. Chemotaxonomic Classification

- ✓ Chemotaxonomic classification combines both taxonomy and chemical constituents for classifying crude drugs.
- ✓ It is based on the presence of characteristic phytochemicals within related plant families or genera.
- ✓ Plants that are taxonomically related often produce similar secondary metabolites such as alkaloids, flavonoids, or essential oils.
- ✓ This method is useful in identifying new medicinal plants, studying plant evolution, and discovering new drugs. It is mainly used in advanced pharmacognosy and research.

Merits (Advantages)

- Highly scientific.
- Combines taxonomy and chemistry.

- Useful in drug discovery.
- Helps identify new medicinal plants.
- Useful for evolutionary studies.
- Supports quality control.
- Valuable in research.

Limitations (Disadvantages)

- Very complex.
- Requires expert knowledge.
- Expensive analytical methods.
- Needs advanced laboratory facilities.
- Not suitable for beginners.
- Time-consuming.
- Limited routine application.

SYPT
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PHARMACY NOTES
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CLASSIFICATION OF CRUDE DRUGS

Systematic arrangement of crude drugs into groups based on common characteristics

QUICK RECALL

1 ALPHABETICAL CLASSIFICATION

- Drugs arranged alphabetically according to English, Latin, scientific or vernacular names.
- Simplest & most common method.

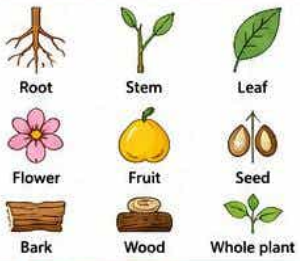


✓ MERITS
- Very simple & easy
- Easy to locate drugs
- Saves time in searching
- Used in pharmacopeias
- No scientific knowledge needed
- Useful for beginners
- Convenient for indexing

✗ LIMITATIONS
- No chemical information
- No therapeutic action
- No botanical relationship
- Similar drugs separated
- Not useful for research
- No idea about morphology
- Cannot compare related drugs

2 MORPHOLOGICAL CLASSIFICATION

- Based on external appearance and plant part used.
- Grouped as roots, stems, leaves, flowers, fruits, seeds, bark, wood or whole plant.
- Useful for identification of organized crude drugs.

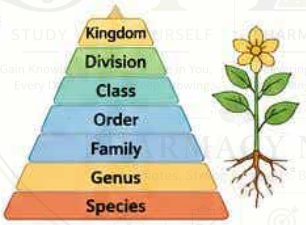


✓ MERITS
- Easy identification
- Based on visible characters
- Useful for practical studies
- Simple classification
- Helps in drug collection
- Useful for organized drugs
- Easy to remember

✗ LIMITATIONS
- Not useful for unorganized drugs
- No information on chemicals
- No pharmacological basis
- Similar parts may differ chemically
- Cannot identify adulteration fully
- No taxonomic relationship
- Limited scientific value

3 TAXONOMICAL CLASSIFICATION

- Based on botanical classification (family, genus, species).
- Shows natural relationship among plants.
- Drugs of same family usually have similar characters.
- Used in botanical & pharmacognostic studies.

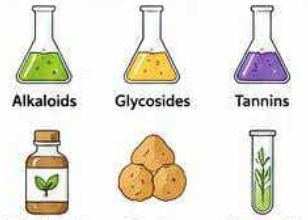


✓ MERITS
- Scientifically accepted
- Shows natural relationship
- Accurate identification
- Useful in botanical studies
- Helps identify related species
- Reduces confusion
- Useful for research

✗ LIMITATIONS
- Difficult for beginners
- No medicinal action
- No chemical classification
- Time-consuming
- Requires botanical knowledge
- Not suitable for animal drugs
- Limited practical use

4 CHEMICAL CLASSIFICATION

- Based on major chemical constituents (alkaloids, glycosides, tannins, volatile oils, resins, flavonoids, steroids, etc.).
- Drugs with similar chemical constituents are grouped.
- Useful for phytochemical studies, QC & isolation of active compounds.

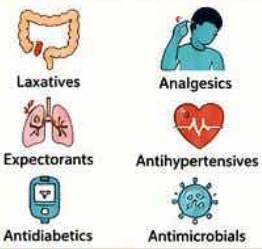


✓ MERITS
- Scientifically reliable
- Useful for phytochemistry
- Helps in drug standardization
- Easy isolation of actives
- Useful for quality control
- Supports research
- Helps in drug development

✗ LIMITATIONS
- Requires chemical analysis
- Expensive
- Complex method
- Some drugs have multiple constituents
- Not useful for beginners
- Ignores botanical relations
- Needs lab facilities

5 PHARMACOLOGICAL CLASSIFICATIONS

- Based on therapeutic effects or pharmacological actions on the human body.
- Drugs with similar medicinal uses are placed in same category (e.g. laxatives, analgesics, expectorants, antihypertensives).
- Useful for doctors, pharmacists & clinicians.

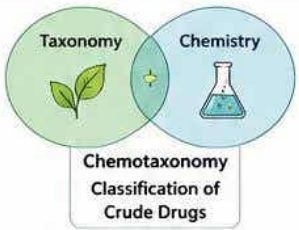


✓ MERITS
- Easy for medical use
- Based on therapeutic action
- Helpful in treatment
- Useful for pharmacists
- Simple to understand
- Helps in prescribing drugs
- Clinically important

✗ LIMITATIONS
- One drug may have many actions
- Similar action from diff. chemicals
- No botanical relationship
- No chemical information
- Difficult for multi-purpose drugs
- Not suitable for research
- Limited scientific classification

6 CHEMOTAXONOMIC CLASSIFICATION

- Combines taxonomy and chemical constituents.
- Based on characteristic phytochemicals in related plant families or genera.
- Useful in identifying new medicinal plants, studying plant evolution & discovering new drugs.



✓ MERITS
- Highly scientific
- Combines taxonomy & chemistry
- Useful in drug discovery
- Helps identify new plants
- Useful for evolutionary studies
- Supports quality control
- Valuable in research

✗ LIMITATIONS
- Very complex
- Requires expert knowledge
- Expensive methods
- Needs advanced lab facilities
- Not suitable for beginners
- Time-consuming
- Limited routine application

QUICK RECALL MNEMONIC